

Metrics for Scientific Discovery: Auditing Embedding Spaces for Novelty, Calibration, and Interpretability

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Abstract

Self-supervised objectives with explicit distributional priors — LeJEPa and its sketched-isotropic-Gaussian regularizer (SIGReg) being the canonical example — promise embedding spaces whose *geometry*, not merely whose linear separability, carries information. This promise matters most in scientific settings, where the object of interest is by construction *unlabeled*: a new particle, a new galaxy morphology, a new material phase. Yet the field evaluates such spaces almost exclusively with linear probes, which measure exactly the one property a discovery pipeline cannot rely on. We argue that a space intended for discovery must be audited in four distinct currencies — *detection*, *calibration*, *discoverability*, and *interpretability* — and we supply a concrete, reproducible battery for each. Our central result concerns the discovery of genuinely new classes: under a specified Gaussian marginal, such a pipeline is not three independent design choices but **one likelihood model at three scales**. The Neyman–Pearson density-ratio functional that shapes the embedding reappears verbatim as the SparKer detection statistic, and the enforced marginal is precisely what makes Euclidean k -means and a unit-variance BIC correctly specified for proposing the new classes — so representation, clustering, and hypothesis test read one quadratic surface. We characterize when this pipeline finds real classes (a *purity gate*) and show that below that gate it is a regularizer wearing a discovery costume. Using a campaign of 79 controlled experiments spanning CIFAR-10/100 and a $4 \times 3 \times 6$ grid of transfer datasets (Stanford Cars, Flowers-102, DTD, Galaxy10 DECals) $\times$ pretraining bases (DINO, LeJEPa, ViSReg) $\times$ contrastive/regularized objectives, we establish four further results. (i) Detection and calibration are *empirically dissociated*: softmax-contrastive spaces win the probe and score exactly 0.000 per-event detection power, while density-ratio-calibrated spaces invert the ordering; no CIFAR-100 space in our sweep exceeds probe 0.94 with tied-Mahalanobis AUC > 0.42 . (ii) The dissociation is *repairable by construction rather than by loss design*: residual fine-tuning, which trains a second head on $\mathbf{z} - \mu_y$ and concatenates, beats the alternative (iterated anchor discovery) in 12/12 dataset $\times$ base cells and yields the first spaces that hold a record probe and calibrated distances simultaneously. (iii) *Scale-free* statistics see novelty that distance-based scores cannot: local intrinsic dimension (LID) reaches AUC = 0.951–0.955 on Flowers-102 using *no novelty labels*, exceeding the best supervised probe (0.906) — the only unsupervised score to beat a trained probe anywhere in the campaign. (iv) Interpretability is a *measurable, dissociable* property: superclass agreement and dendrogram purity track supervision, not probe quality, and a never-labeled held-out class reliably lands among its true semantic siblings. We distill the results into a reporting protocol and a set of regime rules — which objective, which statistic, which repair — for practitioners building embedding spaces meant to find things nobody has labeled.

1 Introduction

Representation learning is evaluated, overwhelmingly, by how well a linear classifier reads a representation out. This is a defensible proxy when the downstream task is classification into known categories. It is the wrong proxy when the downstream task is *discovery*.

A discovery pipeline in the physical sciences — searching for a resonance in collider data, an anomalous transient in a survey, an unfamiliar defect in a microscopy campaign — asks a different question of an embedding. It asks: *given a reference population, is this event unlike anything I have seen, and by how much?* Answering requires the space to support an absolute notion of distance, so that a threshold set on simulated background transfers to data. It requires the tail of the distance distribution to be enriched in genuine novelty, so that clustering the tail recovers new categories rather than mislabeled old ones. And — because a discovery must eventually be *explained* — it requires the geometry to be semantically legible, so a domain expert can say what the new cluster sits next to and why.

None of these are linear separability. All of them are measurable.

The LeJEPA opening The recent line of work that regularizes embeddings toward an explicit distributional target — most directly LeJEPA and its sketched isotropic Gaussian regularizer (SIGReg), which penalizes the sliced 1-D Wasserstein-2 distance between the embedding and $\mathcal{N}(0, I)$ — changes what is available to measure. Once a marginal is *specified* rather than emergent, distances have units: a Mahalanobis radius means something, a density ratio can be calibrated, and a likelihood-based anomaly score becomes well posed. Our starting hypothesis is that this is the correct substrate for discovery, and that its evaluation should exploit the structure it provides instead of discarding it in favor of a probe.

That hypothesis turns out to be half right, in an informative way. A Gaussian marginal is necessary for calibrated geometry but not sufficient for detection, and the two properties trade off sharply enough that we treat them as separate axes throughout.

Contributions

- **A four-currency evaluation battery** (§3) for embedding spaces intended for discovery: detection, calibration, discoverability, interpretability — with concrete estimators, standard protocols, blind spots, and known failure modes for each.
- **Systematic evidence that the currencies dissociate** (§4), across 79 experiments, two image regimes, four transfer datasets, three pretraining bases, and ten training objectives. The dissociation is not noise and is not a tuning artifact: it is reproduced under paired seeds and inverts predictably with the number of seen classes.
- **A unified account of new-class discovery** (§5), which we regard as the central contribution. The Neyman–Pearson functional minimized during representation learning is, verbatim, the SparKer detection statistic evaluated at event rather than pair level; and the SIGReg marginal that licenses the first is what makes the clustering step — Euclidean k -means with a unit-variance BIC — correctly specified rather than merely convenient. We give the *purity gate* that determines when the pipeline finds real classes, show that below it the loop is a regularizer rather than a discovery method (established by a reversal in which the outcome flips sign while purity does not change), and document a bandwidth pathology in kernel two-sample testing that we expect to affect comparisons beyond this work.
- **Two repairs, benchmarked head to head** (§6): iterated anchor *discovery* and *residual fine-tuning*. Residual fine-tuning wins 12/12 cells and produces the campaign’s records.
- **Local intrinsic dimension as a first-class novelty statistic** (§7), including a full control study: holdout rotation, distance baselines, bootstrap intervals, per-class decomposition, and k -sensitivity. It sets a new unsupervised record on Flowers-102 and, importantly, we characterize where it *fails*.
- **Interpretability as a metric, not a figure** (§8): superclass agreement@ k , dendrogram purity, centroid silhouette, and cross-space similarity (CKA, mutual- k NN, Procrustes, ridge R^2 , TwoNN intrinsic dimension), applied to every space in the campaign.

dataset	classes	held out	seen train pool	character
CIFAR-10	10	{4}	45 000	coarse, many samples/class
CIFAR-100	100	{4}	49 500	many-class, low resolution
Stanford Cars	196	186–195	7 712	fine-grained (car models)
Flowers-102	102	92–101	1 840	fine-grained, few-shot (~ 18 /class)
DTD	47	37–46	2 960	texture
Galaxy10 DECals	10	{9}	1 585	coarse morphology, 10%/90% split
FGVC-Aircraft	100	90–99	$\sim 6\,600$	fine-grained (variants)

Table 1: Datasets and open-world splits. Transfer bases are DINO, LeJEPa and ViSReg ViT-B/16 trunks (768-D CLS features); CIFAR uses a ResNet-20 trunk (CIFAR-pretrained, or randomly initialized for the from-scratch lineages).

arm	positives	critic	estimator	marginal	labels
simclr	instance	cosine	softmax	—	no
lejepa	MSE invariance + λ ·SIGReg			global	no
ss (simclr_sigreg)	instance	cosine	softmax	global	no
supcon	supervised	cosine	softmax	—	yes
supcon_sigreg	supervised	cosine	softmax	global	yes
nplm_bilinear	instance	bilinear	NPLM	global	no
nplm_distance	instance	distance	NPLM	global	no
nplm_sup_dist	supervised	distance	NPLM	global	yes
nplm*_cw	(either)	(either)	NPLM	classwise	varies
sup (recipe)	prototype CE to anchors + repulsion			classwise	yes

Table 2: The objectives as coordinates in the design cube. lejepa and the sup recipe are not contrastive and are listed by their components. Standard training: 20 epochs, Adam 10^{-3} , batch 256 two-view (balanced 24/class for classwise arms), $\lambda = 1$.

2 Setup

Open-world protocol In every experiment a set of classes is held out *entirely*: their images and their labels are removed from the training corpus, so at evaluation time they are genuinely novel. Unless noted, the held-out set is the final ten classes of the dataset (Galaxy10: the final one; CIFAR: class 4). Embeddings are then frozen and scored. This is stricter than the common “train-on-everything, test-novelty-by-relabeling” protocol, and we flag the one legacy table (§4.1, aircraft trunk reuse) where an earlier run violated it.

Objectives All objectives are instances of one design cube, which lets us attribute effects to axes rather than to named methods. An objective is specified by four coordinates:

$$(\text{positives}) \times (\text{critic}) \times (\text{estimator}) \times (\text{marginal}).$$

Positives are instance (augmentation) pairs or supervised (same-label) pairs. The *critic* g is cosine $\langle \hat{\mathbf{z}}, \hat{\mathbf{z}}' \rangle / \tau$, bilinear $\langle \mathbf{z}, \mathbf{z}' \rangle / \tau$, or distance $-\frac{1}{2} \|\mathbf{z} - \mathbf{z}'\|^2 / \tau$. The *estimator* is either the standard row-normalized softmax (recovering NT-Xent/SupCon) or the un-normalized maximum-likelihood form

$$\mathcal{L}_{\text{NPLM}} = \mathbb{E}_{\text{ref}}[e^g - 1] - \mathbb{E}_{\text{pos}}[g], \quad (1)$$

whose minimizer is the *absolute* log density ratio $g^* = \log \frac{p(\mathbf{x}, \mathbf{x}')}{p(\mathbf{x})p(\mathbf{x}')}$ with the calibration constraint $\mathbb{E}_{\text{ref}}[e^{g^*}] = 1$. The *marginal* is none, global SIGReg, or classwise SIGReg. The distinction in the estimator axis is the crux of this paper: softmax self-normalizes per row and therefore recovers pointwise mutual information only up to an unknown per-row constant, destroying absolute scale; Eq. 1 does not. We verify the calibration empirically against analytic PMI on a controlled toy ($r > 0.9$).

Two knobs, one inert Two hyperparameters govern the NPLM arms, and they are not symmetric. At initialization the interaction term of Eq. 1 is $\sim 10^{10}$ while $\lambda \cdot \text{SIGReg}$ is $\sim 10^{-2}$: λ is *inert for the first ~ 2 epochs* and only shapes the endgame. By contrast τ is a sharp optimum ($\tau = 1$ for bilinear on CIFAR-100; $\tau \in \{0.5, 2\}$ each cost 0.06–0.10 probe AUC). We therefore scan τ and report λ ablations only where the marginal type changes.

3 A four-currency battery for discovery

We propose that any embedding space advertised for discovery be reported in four currencies. The first two are per-space properties; the third is a property of a *pipeline*; the fourth is a property of the space’s relation to human categories.

3.1 Currency I — Detection

Can novelty be read out at all? Estimator: the AUC of a linear holdout-vs-rest probe trained on frozen embeddings, reported as mean $\pm$ sd over 3 seeds. This is the standard metric and we retain it, with two caveats that recur throughout: single-seed spread is ≈ 0.017 , so **gaps below 0.02 require 3–5 seeds**; and the probe is blind to radial and non-linear novelty structure, which is precisely the structure the other currencies measure.

We also report nearest-centroid top-1 accuracy over seen classes (acc) and the macro one-vs-rest AUC of the prototype posterior (supAUC) as closed-set sanity checks — a space can win the novelty probe while being useless at its nominal task, and we have observed exactly that (nplm-bil-ft on Flowers: probe 0.808 post-discovery, acc 0.531).

3.2 Currency II — Calibration

Do distances have absolute meaning? A discovery threshold is set on a reference population and applied to data; if the score’s scale drifts, the threshold is worthless even when the space is geometrically fine. We estimate calibration at two levels.

Per-space: novelty AUC of the minimum distance to a seen-class centroid (euc1), and of tied and per-class Mahalanobis distances (mahaT, mahaPC; shrinkage 0.1). Plus a Gaussianity summary — eigenvalue conditioning, class RMS, maximum off-diagonal correlation, sliced-Wasserstein ratio, skew and kurtosis — which audits whether the SIGReg target was actually reached rather than merely optimized.

Per-event: the fraction of injected signal events exceeding the $\alpha = 0.05$ null quantile of the reference score distribution. This is the metric that most sharply separates the objective families, and it is the one closest to how a physics search actually operates. **SupCon-family spaces score 0.000 here**, on every dataset, at every signal fraction: a self-normalized objective simply does not produce a transferable threshold.

Dataset-level: three two-sample batteries — SparKer (semi-parametric kernel discrepancy, $M=16$ kernels), Mahalanobis, and MMD — run as toy experiments (N_D events per toy, signal fraction f injected from the holdout pool, power = fraction of 50 signal toys above the $\alpha = 0.05$ quantile of 200 null toys). Two lessons are protocol-critical. First, **kernel statistics must use an annealed median-heuristic bandwidth**: a fixed $\sigma=1$ is calibrated only for the unit-class-width supervised recipe, and a discovery fine-tune that rescales a compact trunk by $\sim 4\times$ sends fixed- σ SparKer to zero power at every fraction *while the space is fine* (annealed: 1.00). We regard this as a general warning about kernel two-sample tests across heterogeneous geometries. Second, statistic and geometry must be matched: parametric statistics need calibrated spaces, and Mahalanobis is simply dead on CIFAR-100 at every dimension and every space (inter-class similarity), where MMD remains the dimension-robust choice.

3.3 Currency III — Discoverability

Does the tail contain the new thing? A discovery loop pools events past a high quantile (0.95) of the seen-distance distribution, selects a cluster count by BIC, merges nearby anchors, and fine-tunes the space on seen labels plus the resulting pseudo-labels. The diagnostic that matters is **pool purity**: the fraction of the pooled tail that is genuinely novel. Everything else about the pipeline is downstream of it.

space (CIFAR-10, 32-D, holdout 4)	probe	acc	eucl	mahaT	per-event
supcon	0.942	0.861	0.689	0.644	0.000
supcon_sigreg	0.882	0.844	0.637	0.577	0.000
sup (supervised SIGReg)	0.923	0.884	0.778	0.645	—
nplm_sup_dist	0.783	0.871	0.726	0.723	0.20
nplm_bilinear (no labels)	0.853	0.546	0.510	0.494	—
simclr	0.867	0.729	0.236	0.229	0.000
lejepa	0.864	0.695	0.408	0.371	—

Table 3: The dissociation in one table. The probe ordering and the calibration ordering are close to reversed. Note `simclr`’s `eucl` of 0.236 — *below* chance-adjusted expectations for a space with a probe of 0.867: the information is there and is not in the distances.

Purity varies over two orders of magnitude across our datasets — 0.795 on DTD, 0.33–0.66 on Flowers, 0.34 on CIFAR-10, 0.03–0.14 on Cars, 0.003–0.013 on CIFAR-100 — and the variation is *structural*, not a tuning failure. On CIFAR-100, 500 holdout images sit inside a ~ 2500 -event tail: the loop is rate-blocked before it begins. On Cars, purity caps at 0.14 for a deeper reason we consider one of the campaign’s cleanest negative results: **a novel car model looks like existing car models**. Pooling requires novelty that is geometrically *outlying*, and fine-grained novelty is not, calibrated space or no. We recommend reporting purity alongside any claimed discovery gain, because a gain at low purity is mechanistically something else (§6).

3.4 Currency IV — Interpretability

Is the geometry legible? Where a superclass partition exists (CIFAR-100 coarse labels, aircraft manufacturer, car make, galaxy morphology) we compute: nearest-class-centroid superclass agreement@1 and @5 against chance, average-linkage dendrogram purity, centroid silhouette, and the within/between distance ratio. Where it does not, we report top-5 nearest-centroid tables, which are directly auditable by a domain expert.

We additionally characterize how spaces relate to *each other*, on aligned samples ($n=4000/\text{cell}$): linear CKA, mutual $k\text{NN}@10$, LLE-weight transfer, orthogonal Procrustes, directional ridge-map R^2 , calibration transfer through a fitted ridge map, and TwoNN intrinsic dimension. This turns “these two objectives learn different things” from an assertion into a measurement, and it is how we establish that the NPLM spaces are *reorganizations* of information the softmax spaces already contain rather than new information (§8).

4 The central dissociation

The single most robust finding across the campaign is that Currency I and Currency II are *anti-correlated across spaces*. Softmax-contrastive objectives maximize linearly readable structure and carry no absolute distance information; density-ratio-calibrated objectives are ready-to-use metric spaces at a probe cost.

On CIFAR-100 the effect is sharp enough to state as a constraint: **no space in the sweep exceeds probe 0.94 with mahaT > 0.42**, across 30+ standalone spaces at 32/64/100-D, hub-initialized and from-scratch. The probe ceiling (≈ 0.95) is owned by supervised softmax and prototype recipes; the mahaT ceiling (≈ 0.47 – 0.49) is owned by NPLM arms. The frontier between them is populated only by *concatenations* (§6).

This is not a seed artifact The one place we initially believed the dissociation was breakable — `nplm_bilinear` reaching probe 0.935 on CIFAR-100, apparent parity with `supcon` — dissolved under paired multi-seed testing: over 5 paired seeds `supcon` gives 0.940 ± 0.004 against `nplm_bilinear`’s 0.855 ± 0.042 , paired difference -0.085 ($t = -4.7$). The single-seed 0.935 was a tail draw from a heavy-tailed seed distribution. The $10\times$ larger seed variance of the label-free NPLM arm is itself an unexplained and, we think, interesting phenomenon.

regime	example	probe champion	calibration champion
few classes, coarse	CIFAR-10, Galaxy10	supcon	nplm_sup_dist
many classes, fine-grained	Cars, Flowers, Aircraft	ss/supcon_sigreg	ss
texture	DTD	simclr	ss
many classes, low-res	CIFAR-100	supcon	nplm_bilinear

Table 4: Regime rules, replicated on all three pretraining bases. Note the inversions: the SIGReg marginal *helps* on fine-grained low-data transfer and *hurts* on CIFAR; label-free instance discrimination wins outright on texture, where labels add nothing to the probe.

The dissociation recurs at every level It is not a property of the loss family alone. It reappears (i) in the fine-tuning objective: prototype/repulsion fine-tuning wins the probe (0.97 vs 0.81–0.84) while NPLM+SIGReg fine-tuning wins *every* power statistic (per-event 0.57 vs 0.30–0.48; SparKer 1.00/0.92; Mahalanobis 0.90; MMD 0.74) and holds round-2 pool purity (0.22–0.30 vs 0.03–0.11); (ii) in the residual construction (§6), where NPLM residuals preserve calibration and plain residuals buy probe with it; and (iii) in semantic structure (§8), where the calibration-strong label-free arms are also the semantically emptiest. The practical rule is uncomfortable but clear: **choose by consumer**. Linear readout $\rightarrow$ prototype/softmax. Detection with a transferable threshold $\rightarrow$ NPLM.

4.1 Regime rules: which corner wins where

The winning objective is determined by regime, and — importantly for practitioners — is *base-invariant*: the same arm wins on DINO, LeJEPa and VISReg trunks. The governing variable is the number of seen classes relative to the embedding dimension, modulated by samples/class.

Three sub-rules are worth isolating.

Anchor realizability gates the classwise marginal. Classwise SIGReg is strictly better than global SIGReg *iff* the class anchors are reachable, i.e. $\dim \geq n_{\text{classes}}$. With 100 anchors in 32-D or 64-D, centroids stall ≈ 3.5 away from their targets and the classwise marginal becomes actively harmful for label-free arms. This is a geometric precondition, checkable before training, and we recommend reporting the centroid-to-anchor distance as a routine diagnostic.

High dimension is not free for calibrated objectives. Moving CIFAR-100 from 32-D to 64-D causes the bilinear NPLM arms to *regress on both currencies* ($0.935 \rightarrow 0.858$; $0.925 \rightarrow 0.825$), while softmax and supervised distance arms gain mildly. On Cars at 100-D the same signature appears from the other side: nplm_sup_dist gains the most probe (+0.096) but loses its per-event calibration lead ($0.106 \rightarrow 0.03$). At high class counts the extra dimensions are spent on probe-readable directions, not calibrated ones.

Fine-tuning inverts the base ranking. On frozen features DINO dominates every currency. After end-to-end fine-tuning the ordering largely dissolves — VISReg holds several pre-discovery records (DTD simclr-ft 0.854, Cars ss-ft 0.741). Cross-space CKA supplies the mechanism (§8): supervision pulls every base toward one shared geometry.

Cross-entropy is not the ceiling. On every substrate tested, a supcon_sigreg head beats a supervised CE head on every column, including — on a LeJEPa fine-tuned trunk — beating CE’s own trained-classifier top-1 with mere nearest-centroid accuracy (0.638 vs 0.625).

5 Discovering new classes: one likelihood model, three scales

We now come to what we regard as the central result of this work. The preceding sections evaluated embedding spaces; this one uses them, and the finding is that the pieces of an open-world discovery pipeline — the representation objective, the clustering that proposes new classes, and the statistical test that certifies them — are not three independent design choices. Under a specified Gaussian marginal they collapse into **one likelihood model instantiated at three scales**, and the places where the pipeline fails are exactly the places where that identification breaks.

5.1 The same Neyman–Pearson functional, twice

The representation objective of §3, written over pairs, is

$$\mathcal{L}_{\text{NPLM}}[g] = \mathbb{E}_{p(x)p(x')}[e^g - 1] - \mathbb{E}_{p(x,x')}[g] \quad (\text{Eq. 1, repeated}). \quad (2)$$

The dataset-level detection statistic we use, SparKer (Grosso et al., arXiv:2511.03095), trains a kernel model f on a data sample D against an anomaly-free reference R by minimizing

$$\mathcal{L}_{\text{NP}}[f] = \sum_{x \in R} w(e^{f(x)} - 1) - \sum_{x \in D} f(x), \quad w = N_D/N_R, \quad (3)$$

and reports $t_{\text{NP}} = -2 \mathcal{L}_{\text{NP}}$, an asymptotic log-likelihood-ratio (Wilks) statistic.

These are the same functional. Eq. (3) is Eq. (1) with pairs replaced by events and the product-of-marginals reference replaced by a control sample. The consequence is not cosmetic: **the objective that shapes the embedding is the objective that later tests it**, so the quantity being estimated during representation learning and the quantity being thresholded at detection time are the same calibrated log density ratio. The $e^f - 1$ term plays an identical role in both — it pins $\mathbb{E}_{\text{ref}}[e^f] = 1$ and thereby fixes the additive constant — which is why a softmax-trained space, whose scale is a gauge choice (App. A), has nothing for a Neyman–Pearson test to threshold and scores 0.000 per-event power.

5.2 Why clustering is correctly specified in a calibrated latent

The third scale is the clustering step, and the same marginal constraint makes it well posed. Our discovery loop selects the number of new classes by BIC under *unit-variance* isotropic Gaussians, i.e. with per-component log likelihood $-\frac{1}{2}\|z - \mu_k\|^2 + \text{const}$ and penalty $k d \log n$. That model is ordinarily a crude approximation. Under SIGReg it is not: once $p(z) = \mathcal{N}(0, I)$ is enforced, Euclidean distance is Mahalanobis distance, so k -means is the correct estimator for this likelihood and the BIC is evaluated under the same density the representation was trained to have. The clustering, the detection statistic, and the loss are all reading the same quadratic surface (§A, final subsection).

This is the constructive payoff of the LeJEPA program that motivated the work. Specifying the marginal is not only a regularizer that improves numbers; it is what licenses the downstream inference machinery to be applied without misspecification.

Where the specification is only partial, and what it costs Two honest caveats. SIGReg constrains the *aggregate* marginal, not the per-class conditionals, so “Euclidean = Mahalanobis” holds globally but not necessarily within a discovered cluster; the classwise marginal repairs this exactly when anchors are realizable (§4.1), which is a checkable precondition and frequently fails. And the unit-variance BIC has a documented pathology at calibrated scale: for a cluster of unit width the $k d \log n$ penalty is smaller than the sum-of-squares gain from splitting, so BIC saturates at k_{max} and *fragments* a single true class. We do not regard this as a bug to be tuned away — it is the correct behaviour of that likelihood — and the loop absorbs it downstream with a weighted union-find merge of anchors closer than a threshold, plus a repulsion exemption between discovered anchors. Fragmentation is nonetheless the mechanism behind the worst outcome we record (*ss-ft/LeJEPA* on Flowers, §5.6).

5.3 The pipeline

Concretely, one discovery round is: (i) embed the unlabeled corpus and score every point by its distance to the nearest seen-class anchor; (ii) *pool* the points past the 0.95 quantile of the seen-population score; (iii) choose k by BIC and cluster the pool; (iv) merge anchors closer than *merge_dist*; (v) assign pseudo-labels and fine-tune the space on seen labels plus pseudo-labels with a prototype/repulsion objective, co-learning the new anchors. Two rounds of five epochs is the settled configuration. The diagnostics recorded per round are the pool size, the **purity** (fraction of the pool that is genuinely novel), $\hat{k}$, the anchor count, the margin AUC of $d_{\text{seen}} - d_{\text{disc}}$, and the per-holdout anchor AUC.

Aircraft, frozen heads, SpK@ $f=0.1$				Cars, e2e ft, SpK@ $f=0.1$	
arm	DINO	LeJEPa	VISReg	arm	DINO
nplm_sup_dist	0.64	0.28	0.68	ss-ft	1.00
supcon	0.42	0.64	0.90	supcon-ft	0.98
supcon_sigreg	0.38	0.52	0.36	nplm-sup-ft	0.94
simclr_sigreg	0.18	0.12	0.10	simclr-ft	0.30
nplm_bilinear	0.16	0.18	0.08	nplm-bil-ft	0.12
lejepa	0.16	0.04	0.06	sigreg-ssl-ft	0.12
nplm_distance	0.14	0.12	0.10		
simclr	0.06	0.44	0.22		

Table 5: SparKer detection power at $\alpha = 0.05$. Aircraft columns use a fixed $\sigma = 1$; all spaces there share the compact frozen-head scale, so comparison *within* the table is fair but not against other tables — the bandwidth caveat above, applied to our own numbers. Cars columns use the annealed schedule.

5.4 SparKer: what it measures and how it must be run

SparKer models the log density ratio as a sparse self-organizing kernel mixture, $f(x) = a^\top [p_{[k]}(x) \cdot k(x)]$ with Gaussian kernels $k_i(x) = \exp(-\|x - \mu_i\|^2 / 2\sigma^2)$ and a gating $p_i = k_i / \sum_j k_j$; both the M kernel centres μ_i (initialized from the data sample) and the weights a are trained on Eq. (3). We use $M = 16$ centres and 300 steps. The bandwidth is *annealed* linearly from σ_0 to $\sigma_0/10$, with t_{NP} recorded at three checkpoints, so a single run probes several resolutions; per- σ p -values are calibrated on anomaly-free toys and aggregated as $-\frac{1}{2} \min_\sigma \log p_\sigma - \frac{1}{2} \overline{\log p_\sigma}$, with detection declared above the $(1 - \alpha)$ quantile of the aggregate’s own null. Power is reported with Clopper–Pearson intervals over 50 signal toys against 200 (pre) or 100 (post) null toys at $\alpha = 0.05$.

The bandwidth rule, and why it generalizes beyond us The single most costly protocol error in the campaign was running SparKer at a fixed $\sigma = 1$. That value is calibrated for the unit class width of the supervised recipe and *only* for it. A discovery fine-tune rescales a compact NPLM trunk by roughly $4\times$; at fixed σ the test then reports zero power at every signal fraction *while the space is perfectly fine* — the annealed schedule scores 1.00 on the same embeddings. We therefore default to the annealed median-heuristic schedule for any comparison across spaces whose geometry may differ, and we flag the affected historical rows rather than silently restating them. The general statement is that a kernel two-sample test carries an implicit length scale, and comparing such tests across representations *silently compares length scales unless the bandwidth is set adaptively*. We suspect this affects published comparisons beyond this work.

5.5 SparKer results

Table 5 gives the picture. **SparKer power is supervised-dominated:** on Cars the three label-bearing arms reach 0.94–1.00 while every label-free arm sits at 0.12–0.30, an eight-fold separation far larger than the corresponding probe gap (0.73 vs 0.59–0.62). Kernel detection is the currency in which supervision matters most, which is the mirror image of DTD, where labels add nothing to the probe.

Two finer points. First, the champion is base-dependent for SparKer (nplm_sup_dist on DINO, supcon on LeJEPa and VISReg) even though it is *not* base-dependent for MMD, where nplm_sup_dist leads on all three (0.86/0.72/0.78). The two kernel statistics disagree about which space is best while agreeing that supervision wins — statistic–geometry matching operating at a finer grain than we can currently explain, and a caution against reporting a single kernel test. Second, on CIFAR-10 SparKer is the statistic that most clearly rewards the discovery step under the corrected bandwidth: the supervised-NPLM classwise space moves $0.42 \rightarrow 0.66$ under an NPLM fine-tune, while the prototype fine-tune that wins the probe moves it $0.42 \rightarrow 0.26$ — the probe/power dissociation of §4, visible in the kernel test.

An acknowledged coverage gap These are the cells for which archived SparKer batteries exist. The residual-fine-tuned and concatenated spaces of §6 — which hold every dataset record in this

dataset	round-1 purity	why	discovery outcome
DTD	0.795 (ss-ft)	textures separate cleanly	rescues weak arms (+0.19)
Flowers	0.33–0.66	coarse-separable holdouts	+0.106 to +0.145
CIFAR-10	0.34	calibrated tail is the novelty	best full pipeline on record
Galaxy10	0.46 (nplm-sup-ft)	coarse morphology	positive 6/6 arms
Aircraft	0.32–0.53	moderate	still probe-negative
Cars	0.03–0.14	novel models look like seen models	probe-flat to negative
CIFAR-100	0.003–0.013	500 holdouts in a 2500-event tail	rate-blocked

Table 6: Purity is necessary and not sufficient. The Cars and CIFAR-100 rows fail for different reasons — geometry and rate respectively — and the Aircraft row fails at adequate purity, which is what forces the refined statement below.

paper — **have no SparKer numbers at all:** the aircraft cells of the end-to-end grid were run with power disabled, and the residual grid never ran power batteries. We have written the audit that closes this (one battery per uncovered cell, loadable from cached embeddings with no retraining) but have not yet run it. Until then the record spaces are characterized by probe, geometry and per-event power only, and any claim that they are also the best *kernel-test* spaces would be unsupported. We flag this rather than quietly reporting the subset that happens to exist.

Statistic–geometry matching No single dataset-level statistic dominates. Parametric statistics (per-event thresholds, Mahalanobis) require the calibrated spaces of §4; kernel statistics (SparKer, MMD) require scale-matched bandwidths. Mahalanobis is simply dead on CIFAR-100 at every dimension and every space, where MMD remains dimension-robust. The practical rule is to report a battery and match the statistic to the geometry, not to pick a favourite.

5.6 The purity gate: when discovery finds classes, and when it does not

Pool purity is the make-or-break diagnostic, and it varies by two orders of magnitude across our datasets for *structural* reasons.

Three conclusions, in increasing order of how much they cost us to learn.

(1) *Pooling requires novelty that is geometrically outlying.* On Cars, purity caps at 0.14 because a novel car model looks like existing car models: the novelty sits *among* the seen classes rather than in the distance tail, and no amount of calibration moves it there. This is the cleanest negative result in the campaign and it bounds the applicability of every tail-pooling discovery method, ours included.

(2) *Above the gate, purity stops being the binding constraint.* Substituting the scale-free LID score (§7) for min-anchor distance as the pool scorer triples round-2 purity — the distance threshold decays as the fine-tune inflates the space, collapsing to 0.18–0.33, while LID holds 0.60–0.68 — and *strictly dominates* as a pooling mechanism. The downstream battery does not move (+0.000, −0.029, +0.007). A mechanism can be strictly better and operationally irrelevant at the same time; only a multi-currency report reveals both facts.

(3) *At low purity, “discovery” is not discovery.* The sharpest evidence is a reversal. On frozen trunks the loop is probe-negative on Cars for every arm; on end-to-end fine-tuned trunks it is probe-positive for all six arms (+0.03 to +0.11) — *with purity below 0.14 in both cases*. Since purity did not change, the gain cannot come from pure pools. The impure-pool prototype fine-tune acts as beneficial refinement when the substrate is aligned to the task and as erosion when it is not. Reported honestly, the low-purity regime is a regularizer wearing a discovery costume, and the largest “gain” in the program is a rescue of a weak head (nplm-sup-ft on DTD, 0.586 → 0.781, replicated on all three bases), not the discovery of anything.

Finally, Aircraft refutes the tempting converse. It pools at adequate purity (0.32–0.53) and discovery *still* costs probe (−0.012 to −0.037) while buying geometry and per-event power (VISReg: per-event 0.30–0.50 across all fractions, mahaT +0.017). Purity is necessary, not sufficient; what fails there is the fine-tune dynamics, not the pool. The corresponding prediction — that freezing the parent half during the discovery fine-tune should make the step non-negative for the first time on

dataset	best construction	probe (per-event)	parent	prior champion
Aircraft/VISReg	supcon→res-nplm cat	0.863 (0.363)	0.809	0.812 [†]
Cars/VISReg	supcon→res-nplm cat	0.855 (0.263)	0.707	0.759
Flowers/DINO	supcon→res-nplm cat	0.885 (0.334)	0.708	0.814
DTD/VISReg	supcon→res cat	0.847 (0.077)	0.750	0.854 [‡]
Galaxy10/LeJEPa	supcon→res cat	0.975 (0.001)	0.919	0.946
CIFAR-100	supcon→res-nplm cat	0.955±0.006	0.944	0.9546
CIFAR-10	supcon→res cat	0.9552±0.0023	0.9404	0.954

Table 7: Best residual construction per dataset (best base shown). Residual fine-tuning beats the discovery pipeline in 12/12 transfer cells. [†]aircraft prior champion carries a holdout-contamination caveat; the row here uses strict open-world parents. [‡]the one exp-70 record residuals do not beat (`simclr-ft` pre-discovery on texture).

fine-grained data — is the one we would test next.

5.7 What this means for a discovery pipeline

Assembling the above: the strongest full pipelines in the campaign are the ones where all three scales are consistent. On CIFAR-10, an NPLM-calibrated space with a supervised distance critic plus prototype discovery reaches post-hoc probe 0.96–0.98 with per-event power 0.73–0.75 against 0.59 for the supervised baseline — the calibrated tail is exactly what pooling needs, the clustering is correctly specified in that tail, and the same NP functional certifies the result. Where any one of the three breaks — rate on CIFAR-100, tail geometry on Cars, fine-tune dynamics on Aircraft — the pipeline degrades in a way the corresponding diagnostic predicts in advance. That predictability, rather than any single AUC, is what we would want carried forward.

6 Repairing the dissociation

Discovery is one of two constructions that might resolve the dissociation of §4. Having characterized it above, we benchmark it head to head against the alternative.

6.1 Discovery as a repair: substrate-dependent and purity-gated

Summarizing §5.6 in the terms of this comparison: across the full grid the discovery loop is probe-positive in 50/72 cells, and the `supcon-ft`+discovery pipeline is positive in 12/12 dataset×base cells — the only universally safe full pipeline we found. Its gains are largest where the pre-discovery space is weakest (+0.05 to +0.15 on the unsupervised arms) and where purity is high, and it is flat or negative on fine-grained data regardless of purity.

Two failure modes bound its use as a repair. Flowers/VISReg shows that purity 0.65+ can still leave a strong arm flat — there was nothing left to learn. And high purity does not protect against a bad *partition*: `ss-ft` discovery on the LeJEPa base collapses the probe 0.792 → 0.613, the worst outcome on record, by splitting round 1 into six anchors at purity 0.605 that fragment the strongest classes rather than the novelty — the BIC fragmentation of §5.2 reaching the probe before `merge_anchors` can absorb it. Treat `ss-ft`+discovery as unstable and prefer `supcon-ft` parents when a discovery step is in the pipeline.

6.2 Residual fine-tuning wins, and breaks the dissociation

The alternative construction: freeze the parent’s seen-class centroids, train a second head end-to-end on the residual $\mathbf{r} = \mathbf{z} - \boldsymbol{\mu}_y$, and evaluate the concatenation $[\mathbf{z}; \mathbf{r}]$. The parent half keeps what supervision learned; the residual half is free to encode the within-class variation that supervision discarded — which is where fine-grained novelty lives.

Three things make this more than a leaderboard entry.

The winning residual objective splits by the same axis as the probe world. NPLM residual concatenation wins on fine-grained data (Cars 3/3, Flowers 2/3); plain softmax residual concatenation wins on texture and coarse data (DTD 3/3, Galaxy10 3/3). The estimator split from §4 reappears

inside the residual construction, which we read as evidence that it is a property of what the objectives encode rather than of how they are deployed.

NPLM residual concatenations keep calibration while raising the probe. Cars/VISReg at probe 0.855 with per-event 0.263 and the best eucl and mahaT on Cars to date is **the first space in the campaign to hold both currencies at once**. Plain residual concatenations, by contrast, often buy their probe *with* calibration — Galaxy10/LeJEPa reaches probe 0.975 at per-event 0.001, which is a warning label, not a win.

The effect is essentially deterministic. Multi-seed checks give Cars/VISReg paired $\Delta = +0.148 \pm 0.004$ (3/3 seeds) and CIFAR-10 paired $\Delta = +0.0157 \pm 0.0004$ (5/5 seeds). We cite records with uncertainties throughout (0.833 ± 0.017 on Cars; 0.955 ± 0.006 on CIFAR-100) because several single-seed peaks in the campaign were fair-but-high draws, and one objective ranking (plain-res vs res-nplm on CIFAR-100) *flipped* under averaging.

Stacking is purity-gated. Running discovery on top of the residual winners stacks only where the residual space already separates the novelty: Flowers $+0.021$ (new record 0.906) and DTD/VISReg $+0.015$ (new record 0.862, finally beating `simclr-ft`), flat or slightly negative elsewhere. On fine-grained data the discovery fine-tune erodes exactly the residual concatenation’s probe directions.

7 Scale-free statistics see what distances miss

Every calibration statistic in §3 is a distance. Distances fail in a specific and predictable way: when novel classes sit *on* the data manifold, near some seen centroid, minimum-distance scores are blind by construction. This is the fine-grained regime — exactly the regime where pooling fails (§6).

We therefore add a *scale-free* statistic: the Levina–Bickel estimator of local intrinsic dimension,

$$\text{LID}_k(\mathbf{x}) = - \left[\frac{1}{k-1} \sum_{j=1}^{k-1} \log \frac{r_j(\mathbf{x})}{r_k(\mathbf{x})} \right]^{-1}, \quad (4)$$

computed against seen-class training references ($k = 20$, max 4000 refs). It depends only on *ratios* of neighbor distances, so it is invariant to the scale deformations that break both fixed- σ kernels and distance thresholds. Higher LID indicates novelty: holdout LID exceeds seen LID in nearly every space we measured — novelty lives in locally higher-dimensional neighborhoods.

Result On Flowers-102, LID reaches $\text{AUC} = 0.951\text{--}0.955$ using *no novelty labels at all*, exceeding the best supervised probe on that dataset (0.906). This is the only unsupervised score to beat a trained probe anywhere in the campaign, and it is a new Flowers novelty record.

Controls We treat a result of this shape with suspicion and ran the corresponding controls. *Holdout rotation*: over 10 random 10-class holdouts on a frozen trunk, Flowers LID gives 0.879 ± 0.018 (min 0.844) versus a $k\text{NN}$ -distance baseline at 0.692 ± 0.047 — classes 92–101 are not special. *Distance baselines*: LID beats $k\text{NN}$ -distance by $+0.08\text{--}0.12$ and eucl by $+0.03\text{--}0.06$ everywhere on Flowers, with Spearman($\text{LID}, k\text{NN}-d$) = $0.55\text{--}0.92$ — correlated, but carrying independent signal. *Robustness*: bootstrap 95% CIs ± 0.006 ; per-holdout-class AUC minimum 0.86–0.92 (no single-class driver); k -sensitivity plateaus at $k \in [20, 50]$ and degrades to ≈ 0.78 at $k = 5$ (the ratio estimator needs neighbors). A rank ensemble with eucl adds nothing — LID subsumes it.

And where it fails The same protocol on Cars gives 0.545 ± 0.040 , and on VISReg fine-tuned Cars spaces the ordering flips: $k\text{NN}$ -distance and eucl beat LID. Aircraft and Galaxy10 sit at 0.56–0.66. LID’s advantage is regime-specific, and the regime is a dataset property, not a protocol artifact — the identical rotation protocol produces 0.88 on Flowers and 0.55 on Cars. The reading: Flowers holdouts sit on the flower manifold with small distances and mediocre distance-based scores, but a distinct local dimensional signature; Cars holdouts do not have one.

LID as a pool scorer Because LID is scale-free it should also fix the failure mode where a discovery fine-tune inflates the space and the distance-based pool threshold decays. It does: substituting LID for min-anchor-distance as the pool scorer holds round-2 purity at 0.60–0.68 where the distance scorer collapses to 0.18–0.33, a $\sim 3\times$ improvement, and *strictly dominates* as a pooling mechanism.

space (CIFAR-100, 100-D)	agree@1	agree@5	dendro purity	silhouette
supcon (parent)	0.700	0.468	0.591	0.197
supcon_sigreg	0.690	0.460	0.603	0.199
supcon→res-nplm (concat)	0.670	0.452	0.564	0.188
supcon→res (concat)	0.640	0.406	0.514	0.162
simclr	0.570	0.366	0.457	0.118
nplm_bilinear	0.530	0.334	0.417	0.059
supcon→res (residual half)	0.490	0.288	0.365	0.029
lejepa	0.450	0.342	0.427	0.046

Table 8: Semantic structure against the CIFAR-100 coarse-label partition (chance agree@1 = 0.040). Semantics track *supervision*, not probe quality and not calibration quality.

But the downstream battery does not move (probe deltas +0.000, −0.029, +0.007). Above the purity gate, purity is not the binding constraint — the fine-tune dynamics are. This is a clean example of why a battery, rather than a single number, is necessary: a mechanism can be strictly better and operationally irrelevant at the same time, and only a multi-currency report shows both facts.

8 Interpretability, measured

A discovery that cannot be explained is not yet a discovery. We treat semantic legibility as a measurable property and find it dissociates from the other currencies as sharply as they dissociate from each other.

The founding question Are otters near beavers? On a CIFAR-100 supervised parent space with beaver held out entirely — never labeled, never in the corpus — the nearest-centroid table reads otter→seal(0.02), beaver*(0.08), bear(0.11) and beaver* →otter(0.08), shrew(0.09), seal(0.10), porcupine(0.13). The never-labeled holdout lands mutually top-2 with otter purely from fine-tuning on the other 99 classes. The same holds on Galaxy10: the never-labeled edge-on bulge sits 0.06 from edge-on no bulge, its true sibling, with cigar smooth next — which is what an edge-on disk looks like. Open-world novelty settles where it semantically belongs, and this is the property that makes a discovered cluster explainable.

Semantics track supervision, not the probe The extreme case is Galaxy10 on DINO: supcon-ft recovers the four-group morphology partition *perfectly* (dendrogram purity 1.000, agree@1 0.9), while the label-free nplm-bil-ft — despite its calibration strengths — has essentially no semantic structure (purity 0.49, silhouette −0.08). The probe-versus-calibration dissociation is *also* a semantic-structure dissociation, and this is the strongest argument in the paper for reporting Currency IV separately: a practitioner selecting on calibration alone can land on a space in which no discovered cluster can be described.

Concatenation is semantically free Residual concatenations inherit the parent’s semantics exactly (Galaxy10 res-nplm concat purity 1.000, equal to parent) while adding the calibration half. On CIFAR-100 the record residual concatenation *tightens* the semantic ring around the held-out beaver (0.06–0.10) at near-parent purity. Meanwhile the residual half *alone* scrambles semantics by design (beaver → mushroom/crab/lobster, purity 0.365) — semantics ride in the parent half, and the concatenation restores them while keeping the probe gain. This is a strong practical argument for concatenation over replacement.

What “fine-grained mistakes” actually are On Cars, the space organizes by body style as much as by make (make-agree@1 0.40 against chance 0.04), with errors like Audi convertible → BMW convertible. These are not failures; they are visually coherent groupings that a make-based partition scores as wrong. We note this because superclass agreement metrics can penalize exactly the structure a scientist would want.

8.1 How spaces relate to each other

Cross-space measurements supply mechanisms for several claims above.

Supervision erases the pretraining base; self-supervision preserves it. The same supervised arm fine-tuned from two *different* bases reaches CKA ≈ 0.74 on average — as similar as two *different supervised losses* on the same base. The label signal pulls every pretraining toward one shared supervised geometry, which is precisely why the regime rules of §4.1 are base-invariant. Label-free `simclr-ft` is far less base-invariant (CKA 0.46–0.62). Consistently, on DTD — where `simclr` wins the probe — `simclr-ft`’s *local* structure transfers across bases better than the supervised arms’ (mutual- k NN 0.28 vs 0.16): its texture neighborhoods are the transferable part.

NPLM spaces are reorganizations, not new information. The NPLM arms are geometric outliers everywhere (CKA 0.2–0.36, mutual- k NN 0.03–0.06 against every other arm, LLE weight-transfer ratio > 1) — yet they are highly *decodable* from softmax spaces (asymmetric ridge R^2 up to 0.97 at CKA 0.36). Calibration is a re-encoding of information the supervised space already contains. This explains why concatenation works so well and why a single loss struggles: the two currencies are different *codes* for overlapping content.

Residual children are mostly linear re-maps — and drift is capacity, not help. Parent-vs-child ridge R^2 is 0.97–0.98 on coarse data (the child is nearly a linear re-map; calibration is re-weighting) and 0.91–0.95 with mutual- k NN ≈ 0.2 on fine-grained data (locally reorganized, 5–10% genuinely new variance). Within the transfer grid the concatenation gains track the unexplained fraction. But CIFAR refutes the naive version of this rule: there children drift *much* further (R^2 down to 0.32) while producing the campaign’s *smallest* probe gains (+0.006–0.016 vs Cars’ +0.148). Unexplained variance is necessary and not sufficient — what pays is whether the new variance encodes the holdout.

Loss families have intrinsic-dimension fingerprints. TwoNN intrinsic dimension orders the arms identically in every cell: NPLM supervised most compressed (ID ≈ 2 –3), SIGReg/SSL arms mid (5–7), softmax and frozen highest (9–13). The Gaussian target and the density-ratio collapse are directly visible as dimension, which makes ID a cheap objective-provenance diagnostic.

9 A reporting protocol

We distill the campaign into a protocol we would want applied to any embedding space advertised for scientific discovery.

1. **Report all four currencies.** Probe-only or geometry-only conclusions are, in our experience, wrong roughly as often as they are right, because the currencies are anti-correlated by construction.
2. **State the number of seen classes, the embedding dimension, and the centroid-to-anchor distance.** These determine which corner of the design cube is even available (classwise marginals require $\text{dim} \geq n_{\text{classes}}$).
3. **Never claim a probe gap below 0.02 from one seed.** Single-seed spread is ≈ 0.017 . Two rankings in this campaign flipped under averaging.
4. **Use annealed median-heuristic bandwidths** for any kernel two-sample statistic compared across spaces with different scales. A fixed bandwidth reports zero power on a space that is fine.
5. **Report pool purity next to any discovery gain.** A gain at purity < 0.15 is refinement, not discovery, and behaves differently (it depends on substrate alignment, not on the pool).
6. **Include at least one scale-free statistic.** Distance scores are structurally blind to on-manifold novelty; LID costs nothing and, in one regime, beat every supervised alternative we had.
7. **Audit semantics explicitly.** Superclass agreement, dendrogram purity, and a nearest-centroid table a domain expert can read. Calibration does not imply legibility — we have a space with strong calibration and silhouette -0.08 .

And a default recipe For a practitioner who wants one starting point: fine-tune end-to-end with `supcon` (adding a global SIGReg marginal if the data are fine-grained and low-shot), then train a

residual head on $\mathbf{z} - \mu_y$ with an NPLM objective on fine-grained data or a softmax objective on coarse/texture data, and evaluate the concatenation. Add a discovery pass only if measured pool purity exceeds ≈ 0.3 . Score with the four-currency battery, including LID.

10 Limitations

Our evidence is entirely in the image domain; the regime rules are stated in terms of class counts and samples per class and we expect them to transfer, but we have not tested them on the scientific modalities (event-level collider data, spectra, time series) that motivate the work. The “discovery” setting is simulated by holding out known classes, which is a proxy: real novelty is not a withheld member of a curated taxonomy, and the Cars result — where novel classes are not geometrically outlying — suggests the proxy’s difficulty varies more than the datasets’ nominal difficulty does. Several conclusions rest on 3–5 seeds, which resolves 0.02-scale effects but not smaller ones. Our SparKer coverage is incomplete in exactly the place it matters most: the residual and concatenated spaces that hold every record here have no kernel-test numbers (§5.5), so the claim that one construction serves both currencies rests on per-event power rather than on the full detection battery. Our discovery loop is one specific algorithm (quantile pool, BIC k -selection, anchor merge, prototype fine-tune) and the purity-gating conclusion may be specific to it. Finally, the interpretability metrics require a superclass partition, which is available in our datasets and frequently is not in the settings we care about; the nearest-centroid tables remain available but require expert reading rather than a number.

11 Conclusion

Embedding spaces built for scientific discovery need to be evaluated as instruments, not as classifiers. When we do so — across 79 controlled experiments, seven datasets, three pretraining bases and ten objectives — the picture is consistent. The part we would most want carried forward is the unification of §5: a pipeline that discovers new classes need not bolt a clustering heuristic and an anomaly test onto a representation trained for something else. Fix the marginal, and the density-ratio objective that shapes the space is the statistic that certifies a discovery, while the clustering that proposes candidates is correctly specified under the same density. When the pipeline then fails, it fails for a reason the diagnostics name in advance — rate on CIFAR-100, tail geometry on Cars, fine-tune dynamics on Aircraft — which we value more than any single AUC. Alongside that: linear readability and calibrated geometry are anti-correlated properties that no single loss in our design space achieves together; the repair is architectural (residual decomposition and concatenation) rather than a better loss; scale-free statistics reach novelty that distance-based scores cannot see, in a regime one can identify in advance; and semantic legibility is a third, separately measurable axis that tracks supervision alone. The distributional-prior program that LeJEPa and SIGReg opened up is, we think, the right foundation — specifying the marginal is what makes any of these quantities well posed. What it needs alongside it is a metrology: a standard battery, honest uncertainties, and the discipline to report the currency a space *lacks*.

Reproducibility statement

All experiments are scripted (`experiments/NN_*.py`, each with a mandatory `--quick` smoke mode) over a shared library (`supersig/`: losses, training loops, metrics, discovery). Result arrays are archived as `logs/expNN/*.npz` with fixed key conventions, condensed into per-dataset master tables under `docs/`. Seeds are fixed and identical across paired cells; the probe uses seeds $1000+s$. Known reproduction jitter from cuDNN nondeterminism is $\approx \pm 0.003$ in AUC and is stated wherever single-seed records are cited.

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A Gradient descent under NPLM versus InfoNCE, and the log-likelihood readout

The dissociation of §4 is stated empirically in the main text. This appendix derives it. We compare the two estimators of the same population object — the pointwise mutual information — at the level of their *gradients*, and show that essentially every qualitative difference reported in the campaign (the vanishing per-event power of softmax spaces, the inertness of λ , the heavy-tailed seed variance of the bilinear NPLM arm, the base-robustness of the distance NPLM arm, and the intrinsic-dimension fingerprints) follows from a single structural fact: **InfoNCE is flat along the direction that carries the absolute normalization, and NPLM is strictly convex along it.**

We flag scope honestly. The derivations in §A.2–§A.4 are analytical and new to this paper; the links drawn to campaign measurements in §A.5 are *post-hoc consistency checks* against data already collected, not preregistered tests. We indicate in each case what a decisive experiment would be.

A.1 Two estimators, one target

Fix a batch $\{(x_i, x_i^+)\}_{i=1}^N$ and write $g_{ij} = g_\theta(x_i, x_j^+)$, with the diagonal as positive pairs and the off-diagonal as the shuffled product-of-marginals reference. The two objectives are

$$\hat{\mathcal{L}}_{\text{NCE}} = -\frac{1}{N} \sum_i \left[g_{ii} - \log \sum_j e^{g_{ij}} \right], \quad (5)$$

$$\hat{\mathcal{L}}_{\text{NPLM}} = \frac{1}{N(N-1)} \sum_{i \neq j} (e^{g_{ij}} - 1) - \frac{1}{N} \sum_i g_{ii}. \quad (6)$$

They share the batch geometry exactly; only the functional differs. Both are consistent estimators of PMI — but of different *gauge classes* of it, as the gradients make immediate.

A.2 The row-sum identity: a flat direction versus a convex one

Differentiating (5) with respect to the critic entries gives

$$\frac{\partial \hat{\mathcal{L}}_{\text{NCE}}}{\partial g_{ij}} = \frac{1}{N} (\sigma_{ij} - \delta_{ij}), \quad \sigma_{ij} = \frac{e^{g_{ij}}}{\sum_k e^{g_{ik}}}, \quad (7)$$

while (6) gives

$$\frac{\partial \hat{\mathcal{L}}_{\text{NPLM}}}{\partial g_{ij}} = \begin{cases} \frac{e^{g_{ij}}}{N(N-1)}, & i \neq j \text{ (reference)}, \\ -\frac{1}{N}, & i = j \text{ (positive)}. \end{cases} \quad (8)$$

Three differences are visible at once.

(i) InfoNCE gradients are row-stochastic; NPLM’s are not From (7), $\sum_j \partial \hat{\mathcal{L}}_{\text{NCE}} / \partial g_{ij} = 0$ for *every* anchor i , because σ_i is a probability vector. The InfoNCE gradient is therefore purely *relative*: it can reallocate mass among the entries of a row but can never change a row’s total. Formally, consider shifting the critic by an arbitrary per-anchor function, $g_{ij} \mapsto g_{ij} + c_i$. Then σ_{ij} is unchanged and

$$\hat{\mathcal{L}}_{\text{NCE}}[g + c] = \hat{\mathcal{L}}_{\text{NCE}}[g] \quad \text{exactly, for all } c. \quad (9)$$

The per-anchor shift is an exact *flat direction* of the loss — zero gradient, zero curvature. This is the entire content of “InfoNCE recovers PMI up to a per-row constant”: the constant is not merely unidentified, it is *invisible to gradient descent*, so no amount of training, data, or capacity can recover it.

For NPLM the same shift gives, with $\bar{c}$ a uniform shift for clarity,

$$\hat{\mathcal{L}}_{\text{NPLM}}[g + \bar{c}] = e^{\bar{c}} \hat{\mathbb{E}}_{\text{ref}}[e^g] - 1 - \hat{\mathbb{E}}_{\text{pos}}[g] - \bar{c}, \quad (10)$$

so that

$$\frac{d \hat{\mathcal{L}}_{\text{NPLM}}}{d \bar{c}} = e^{\bar{c}} \hat{\mathbb{E}}_{\text{ref}}[e^g] - 1, \quad \frac{d^2 \hat{\mathcal{L}}_{\text{NPLM}}}{d \bar{c}^2} = e^{\bar{c}} \hat{\mathbb{E}}_{\text{ref}}[e^g] > 0. \quad (11)$$

The direction InfoNCE cannot see is *strictly convex* for NPLM, with a unique minimum at $\bar{c}^* = -\log \hat{\mathbb{E}}_{\text{ref}}[e^g]$, i.e. exactly at the calibration condition $\hat{\mathbb{E}}_{\text{ref}}[e^{g^*}] = 1$. Equivalently: summing (8) over all pairs,

$$\sum_{i,j} \frac{\partial \hat{\mathcal{L}}_{\text{NPLM}}}{\partial g_{ij}} = \hat{\mathbb{E}}_{\text{ref}}[e^g] - 1 = (\text{calibration residual}). \quad (12)$$

The total gradient of the NPLM loss is the miscalibration. Gradient descent on (6) performs calibration as an inseparable part of fitting, and (12) doubles as a free training diagnostic: a run whose summed critic gradient has not approached zero has not calibrated, whatever its loss value.

This is the precise sense in which per-event detection power is *structurally* unavailable to a softmax objective. A per-event threshold is a statement about the absolute value of the score; (7) says that value is a gauge choice on which the objective exerts no force. The observed 0.000 per-event power of every SupCon-family space in the campaign is not a tuning failure or a weak baseline — it is this identity.

(ii) Why max-subtraction destroys calibration The standard numerical guard for softmax losses, $g_{ij} \mapsto g_{ij} - \max_k g_{ik}$, is a per-anchor shift. By (i) it is exactly free for InfoNCE and exactly *fatal* for NPLM: it moves the minimizer from g^* to $g^* + c$, the one quantity the loss was chosen to pin. The correct guard clamps the exponent, $e^{\min(g, g_{\max})}$, which by (8) sets the gradient of pathological reference pairs to a constant instead of shifting the optimum, leaving g^* untouched wherever $g < g_{\max}$. The campaign uses $g_{\max} = 30$ and verifies the resulting critic against analytic PMI at correlation > 0.9 with mean offset ≈ 0 ; max-subtraction drives the offset to hundreds.

(iii) Bounded versus exponential gradients From (7), $|\partial \hat{\mathcal{L}}_{\text{NCE}} / \partial g_{ij}| \leq 1/N$ *unconditionally*. InfoNCE cannot produce a large critic gradient no matter how badly scaled the embedding is. From (8), the NPLM reference gradient is $e^{g_{ij}} / N(N-1)$, which is unbounded above. This single asymmetry accounts for the optimization folklore around NPLM, and we can quantify it.

A.3 Gradient variance and the two critics

Suppose the critic values on reference pairs are approximately Gaussian, $g \sim \mathcal{N}(\mu, s^2)$. Then the relative variance of the reference gradient is

$$\frac{\text{Var}[e^g]}{(\mathbb{E}[e^g])^2} = e^{s^2} - 1, \quad (13)$$

which grows *exponentially* in the spread of the critic, while the corresponding quantity for (7) is $O(1)$ because $\sigma_{ij} \in [0, 1]$. A modest critic spread of $s = 3$ already gives a relative gradient standard deviation of ≈ 90 . NPLM training is therefore not merely “sometimes unstable”: its gradient noise is controlled by a quantity, s , that is itself a free consequence of the parametrization.

This makes the *critic* axis of the design cube, which in the main text is one knob among four, the decisive one for optimization. Compare the two critics used in the campaign:

$$\text{distance: } g = -\frac{1}{2} \|z - z'\|^2 / \tau \leq 0 \quad \implies \quad e^g \leq 1 \quad (\text{bounded by construction}), \quad (14)$$

$$\text{bilinear: } g = \langle z, z' \rangle / \tau \in \mathbb{R} \quad \implies \quad e^g \text{ unbounded.} \quad (15)$$

The distance critic makes the exponential gradient *automatically bounded* — the pathology of (13) cannot occur, and the clamp never activates in the normal range. The bilinear critic has no such protection: a single well-aligned reference pair can dominate the batch gradient.

A.4 The force law on the embedding

Push the gradients through to the embedding itself. Under the distance critic (the form the critic collapses to once SIGReg is active), $\partial g_{ij} / \partial z_i = -(z_i - z_j^+)$, and

$$\nabla_{z_i} \hat{\mathcal{L}}_{\text{NPLM}} = \underbrace{\frac{1}{N} (z_i - z_i^+)}_{\text{spring to positive}} - \underbrace{\frac{1}{N(N-1)} \sum_{j \neq i} e^{-\frac{1}{2} \|z_i - z_j^+\|^2} (z_i - z_j^+)}_{\text{repulsion}}. \quad (16)$$

	InfoNCE / SimCLR	NPLM
positive-pair gradient	$\frac{1}{N}(\sigma_{ii} - 1)$, saturating	$-\frac{1}{N}$, constant
reference-pair gradient	$\frac{1}{N}\sigma_{ij} \in [0, \frac{1}{N}]$	$e^{g_{ij}}/N(N-1)$, unbounded
row sum of gradients	0 (exactly)	calibration residual, Eq. (12)
per-anchor shift $g \rightarrow g + c$	flat direction	strictly convex, Eq. (11)
recovered object	PMI + $c(x)$	PMI, absolute
repulsion range	long (rank/softmax)	short (Gaussian kernel)
relative gradient variance	$O(1)$	$e^{s^2} - 1$
stabilizer	max-subtraction (free)	exponent clamp (max-sub. is fatal)

Table 9: Gradient properties of the two estimators of the same population target. Every entry follows from (7)–(8).

Two features of (16) are worth stating plainly.

The repulsion is the gradient of a kernel density estimate. The weight $e^{-\frac{1}{2}\|z_i - z_j\|^2}$ is an unnormalized isotropic Gaussian kernel, so the second term is precisely $\nabla_{z_i} \hat{q}(z_i)$ for $\hat{q}$ a Gaussian KDE of the reference population. NPLM gradient descent is therefore: *pull toward your positive with a linear spring, and descend the estimated reference density.* The unit-width Gaussian supplies an intrinsic **length scale**, and that length scale is what makes distances absolute. A point far from all references feels essentially *no* repulsion — the force vanishes as $e^{-d^2/2}$ — so the space does not expend capacity separating what is already separated.

InfoNCE has no length scale. The corresponding repulsion weight is σ_{ij} , which is normalized *within a row*: even if every negative is far away, the row still sums to one and the repulsion is redistributed rather than extinguished. Contrastive repulsion is thus long-range and rank-based, and its fixed point is uniformity on the sphere rather than any particular scale. This is the geometric statement behind the main text’s observation that `simclr` attains probe 0.867 with `eucl` 0.236: the objective arranges *angles* correctly and never had an opinion about radii.

The positive term does not saturate. In (7) the positive gradient is $\frac{1}{N}(\sigma_{ii} - 1)$, which $\rightarrow 0$ as the positive is separated: InfoNCE anneals itself. In (8) it is the constant $-\frac{1}{N}$. NPLM pulls positives together with undiminished force forever, and equilibrium is reached *only* when the exponential repulsion balances it — that is, at the calibration condition, not at a separation criterion.

A.5 What this predicts, and what the campaign measured

The derivation yields five predictions. All five have a counterpart in the campaign data. We stress again that these are consistency checks on existing measurements, not independent confirmations.

1. *Softmax spaces have exactly zero transferable per-event power, regardless of probe quality.* Predicted by the flat direction in (i). Measured: per-event 0.000 for every SupCon-family space on every dataset and every signal fraction, including spaces at probe 0.94.
2. *λ is inert early; τ is not.* At initialization with a bilinear critic on raw (unnormalized) embeddings, g has spread of order tens, so by (8) the interaction gradient is of order e^g — numerically $\sim 10^{10}$ against $\sim 10^{-2}$ for $\lambda \cdot \text{SIGReg}$. The marginal term cannot compete until the exponent has contracted, which takes ≈ 2 epochs; λ then shapes only the endgame. By contrast τ rescales the exponent itself and hence enters the gradient multiplicatively at all times. Measured: λ ablations are flat, $\tau = 1$ is a sharp optimum on CIFAR-100 with $\tau \in \{0.5, 2\}$ each costing 0.06–0.10 probe.
3. *The bilinear NPLM arm should be the high-variance one; the distance NPLM arm should be the stable one.* Predicted by §A.3: $e^g \leq 1$ automatically for the distance critic, unbounded for bilinear. Measured, and to our mind the appendix’s most striking correspondence: `nplm.bilinear` on CIFAR-100 gives 0.855 ± 0.042 over paired seeds, a spread $10\times$ that of `supcon` (0.940 ± 0.004), with a heavy right tail that produced a single-seed 0.935 we initially mistook for parity; whereas `nplm.sup_dist` is *the most base-robust arm in the campaign*,

probe 0.72–0.73 across all three pretraining trunks (spread 0.009, versus supcon’s 0.076). The unexplained seed variance flagged in §4 is, on this account, the exponential gradient variance of (13) — and the prediction is that it is a property of the *critic*, not of the estimator. A decisive test is cheap and we have not run it: hold the estimator at NPLM, sweep the critic between distance and bilinear at matched τ , and compare seed variance directly.

4. *NPLM spaces should be compressed relative to softmax spaces.* Predicted by the short-range force law: distant pairs exert no force, and the non-saturating positive pull is balanced only at the calibration scale. Measured: TwoNN intrinsic dimension orders the arms identically in every cell of the grid — supervised NPLM $\approx$ 2–3, SIGReg/SSL 5–7, softmax and frozen 9–13.
5. *NPLM should reorganize rather than add information.* The Bregman identity for (6) states that the constrained minimizer is the projection of PMI onto the critic family in the divergence generated by $\phi(t) = e^t$; a different projection of the same target is still a function of that target. Measured: NPLM spaces are geometric outliers against every other arm (CKA 0.2–0.36, mutual- k NN 0.03–0.06) yet are highly *decodable from* softmax spaces (ridge R^2 up to 0.97 at CKA 0.36).

A.6 Why the log-likelihood readout needs both terms

Finally, the gradient view sharpens why neither objective alone delivers an interpretable likelihood surface, which is the motivation for the hybrid used throughout the campaign.

NPLM fixes the interaction and its absolute scale, but the critic family $g = \langle z, z' \rangle + b(x) + b(x')$ retains a gauge,

$$z \mapsto Rz + a, \quad b(x) \mapsto b(x) - \langle a, z \rangle - \frac{1}{2}\|a\|^2 + \beta, \quad (17)$$

under which g , and hence the loss and *every gradient*, is exactly invariant. So NPLM alone determines what the pair interaction is but not how it is split between the embedding and the bias — the map into “log-likelihood space” is undetermined by descent. SIGReg removes precisely this residual freedom: pinning $p(z) = \mathcal{N}(0, I)$ kills the shift a (leaving only an irrelevant rotation R) and identifies the bias as the marginal log-density, $b_\theta(x) \rightarrow -\frac{1}{2}\|z\|^2 + \text{const}$. Substituting back collapses the critic to the pure distance form and yields the three quadratic identities

$$\log p(x) = -\frac{1}{2}\|z\|^2 + c_0, \quad \log \frac{p(x, x')}{p(x)p(x')} = \langle z, z' \rangle, \quad \log p(x, x') = -\frac{1}{2}\|z - z'\|^2 + c_1. \quad (18)$$

The division of labour is exact and is a statement about descent directions: **NPLM supplies curvature along the normalization direction that InfoNCE leaves flat; SIGReg supplies curvature along the gauge directions that NPLM leaves flat.** Each term is the other’s null space.

Two caveats bound the claim, and both are visible in the main text’s results. The identities hold exactly only when PMI lies in the bilinear-plus-separable family; otherwise the Bregman projection returns the nearest such surface, and the residual is real — it is one reading of why concatenating a softmax half with an NPLM half outperforms either alone (§6), since the two halves are different projections of overlapping content. And the readout degrades under collapse: at small embedding dimension and small τ the supervised optimum is a simplex equiangular tight frame in which all class centres are equidistant and the likelihood information is destroyed. The campaign’s realizability condition ($\dim \geq n_{\text{classes}}$ for classwise anchors, §4.1) is the practical form of that caveat, and the observed regression of the bilinear NPLM arms at 64-D on CIFAR-100 — losing *both* currencies — is the kind of failure it predicts.